



Designed by Hieronymus Fabricius, the dissection theater in Padua, Italy, offered public demonstrations of anatomical dissections.

THE UNIVERSITY OF PADUA

had been at the forefront of the “golden age” of anatomy since Andreas Vesalius became professor of surgery and anatomy there in 1537. The university attracted students from all over Europe, and the department was led by a succession of distinguished surgeons and anatomists.

Hieronymus Fabricius was appointed to the post in 1565, and became well known for demonstrating **the dissection of both humans and animals**, and instituting a new style of investigative anatomy. In order to make these demonstrations available to a wider audience, he designed a theater for dissections. The **theater was built** in 1594 with funds provided by the Senate of the Republic

of Venice. Although some public dissections had been performed before, this was the first **permanent structure designed and built especially for such demonstrations**. He was succeeded by his students, Julius Casserius, and later Adriaan van der Spiegel, who continued the tradition of public demonstrations of anatomical dissections.

In the same year, Simon Stevin wrote his treatise, *Arithmetica*. This book dealt with, among other things, the **solution of quadratic equations**—equations involving a squared quantity—and important concepts in the field of **number theory**.

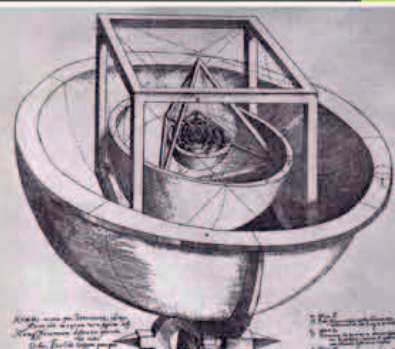
HIERONYMUS FABRICIUS (1537–1619)



Born in Acquapendente, Italy, Hieronymus Fabricius studied at the University of Padua, where he eventually became Professor of Anatomy in 1562 and Professor of Surgery in 1565. Famous for his public demonstrations of anatomical dissections, he is best known as a pioneer in the field of embryology and for describing the valves in veins.

“ I MUCH PREFER THE SHARPEST CRITICISM OF A SINGLE INTELLIGENT MAN TO THE THOUGHTLESS APPROVAL OF THE MASSES... ”

Johannes Kepler, German astronomer



An illustration of Kepler's planetary model from *Mysterium Cosmographicum*.

IN 1596, GERMAN ASTRONOMER

JOHANNES KEPLER (1571–1630) published his first important work on astronomy, *Mysterium Cosmographicum* (*The Cosmographic Mystery*). As well as defending the heliocentric model of the universe proposed by Copernicus (see 1543), Kepler **explained the orbits of the known planets around the Sun in geometric terms** in an attempt to unravel “God’s mysterious plan of the universe.” To do this, he drew upon the classical notion of “the harmony of the spheres,” which he linked to the five Platonic solids—octahedron, icosahedron, dodecahedron, tetrahedron, and cube. These, when inscribed in spheres and nested inside one another in order, corresponded to the orbits of the planets Mercury, Venus, Earth, Mars, Jupiter, and Saturn.

In 1596, Flemish cartographer **Abraham Ortelius** noted that the coastlines on either side of the Atlantic Ocean seemed to fit like pieces of a puzzle. He was the first to suggest that Africa, Europe, and the Americas may once have been connected. Although he attributed their separation to a major cataclysm, his ideas anticipated the modern theory of continental drift (see 1914–15).

Also in 1596, English author **Sir John Harington** (1561–1612) published *A New Discourse*



upon a Stale Subject: The Metamorphosis of Ajax. This text was part political satire and part description of his invention, a **rudimentary flush toilet** called “Ajax.” The name was a pun on “a jakes,” contemporary slang for toilet. The invention was a major step toward modern sanitation.

Water closet

John Harington's “Ajax,” the prototype of the modern flush toilet, was invented with the aim of eliminating disease.

One of the most important textbooks of alchemy, *Alchemia*, was published in 1597 by German metallurgist **Andreas Libavius**. Unlike previous books on alchemy, *Alchemia* stressed the importance of systematic laboratory procedures. It also contained a catalog of various medicaments and metals, and included the first description of the properties of zinc.

Zinc

Known in China and India since the 14th century, zinc was first described in Europe by 16th-century alchemist **Andreas Libavius**.



1594

Hieronymus Fabricius opens the first permanent theater for public anatomical dissections in Padua

1596 Johannes Kepler publishes *Mysterium Cosmographicum*, defending the Copernican heliocentric system of planetary motion

1596 John Harington describes his invention of an early flush toilet, the “Ajax”

1596 Austrian astronomer and mathematician Georg Rheticus's trigonometric tables are published posthumously

1594 Simon Stevin publishes *Arithmetica*

1596 Abraham Ortelius compares the coastlines of continents and suggests that they may have been joined at one time

1597 Andreas Libavius publishes *Alchemia*